

Islam Kathat

Software Engineer, AI/ML | Python, GenAI, Deep Learning, LLM Agents & RAG

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SUMMARY

Software engineer with 1 year of professional experience in Python, GenAI, machine learning, and deep learning: production tool-calling agents, RAG pipelines, LLM fine-tuning, and the batch pipelines and APIs behind them. Pretrained a 350M-parameter GPT end to end on 6.83B tokens (data pipeline, GPU training loop, profiling, crash recovery), then ran SFT and QLoRA post-training. Covers the model lifecycle from training to deployment.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript, JavaScript, Bash

LLMs & Agents: GenAI, large language models (LLMs), foundation models, AI agents, tool calling, agent orchestration, RAG, prompt engineering, embeddings, vector database (ChromaDB), hybrid retrieval, cross-encoder re-ranking, LLM evaluation, responsible AI

ML & Training: PyTorch, torch.compile, CUDA, Transformers, TRL, PEFT, Accelerate, BitsAndBytes, Flash Attention, mixed precision (BF16/FP16), gradient checkpointing, distributed and multi-GPU training, supervised fine-tuning, LoRA/QLoRA, NF4 quantization, KV cache, NumPy, Pandas, Scikit-learn

Infrastructure & Tooling: Docker, Redis, RabbitMQ, Kafka, AWS S3, REST APIs, FastAPI, PostgreSQL, MongoDB, cloud storage, Weights & Biases (experiment tracking and monitoring), Hugging Face Hub, Git, pytest, Linux, profiling and performance debugging, React, Next.js

EXPERIENCE

JSB Global Infotech

May 2025 – May 2026

Full-Stack AI Engineer

Jaipur, India

- Built and operated scheduled batch AI pipelines (FastAPI, Node.js, RabbitMQ, Redis, Docker, AWS S3) that scraped, rewrote with AI, and published content end to end with no manual steps.
- Fine-tuned and deployed a retrieval-grounded (RAG) LLaMA 3.1 support assistant on the company site, replacing templated replies with answers grounded in the company's own documents.
- Cut API response time 35% by rewriting MongoDB aggregation pipelines and adding Redis caching across React Native (Expo) and Next.js products serving live users.

CodeAlpha

Jul 2024 – Aug 2024

Data Science Intern

Remote

- Rebuilt classification feature engineering and hyperparameter search (Scikit-learn, PyTorch), raising held-out accuracy 15%.

PROJECTS

350M-Parameter GPT: Pretraining, Post-Training & Training Infrastructure

[GitHub](#) | [Model](#) | [Training logs](#)

PyTorch, CUDA, torch.compile, Flash Attention, BF16, Hugging Face, Weights & Biases

- Trained a decoder-only transformer (24 layers, 16 heads, 1024 dim, 50,304 vocab, RMSNorm, Flash Attention, weight tying) on 6.83B tokens to validation loss 3.04 and perplexity 20.9, comparable to GPT-2 XL at 6x fewer tokens.
- Diagnosed a recurring torch.compile and BF16 out-of-memory failure (AOT Autograd materializing 12GB FP32 gradient buffers) and excluded lm_head and cross_entropy from the compiled graph via torch.compiler.disable, restoring training on the same hardware.
- Built a fault-tolerant, resumable data and checkpointing pipeline: tokenized the corpus on free Colab CPU into uint16 shards, shipped them to rented RunPod GPUs with rclone so paid GPU time went to training only.
- Ran supervised fine-tuning on OpenHermes 2.5 (746K examples, 288M tokens) with identity-protection examples to prevent factual bleeding; tracked every run in Weights & Biases and shipped to Hugging Face.

ParcelPilot AI Support Agent: Tool-Calling Agent Runtime & Evaluation Harness

[Live](#) | [Backend](#) | [Frontend](#)

Python, FastAPI, streaming tool-calling loop, hybrid retrieval, pytest, React, TypeScript

- Wrote the agent runtime rather than adopting a framework: a tool-calling loop with 7 tools, a bounded 6-step budget, and per-index accumulation of fragmented argument deltas; every call and token streamed live.
- Built agent governance into the tool layer, not the prompt: the caller's principal is injected server-side and never exposed as a model-writable argument, so cross-account reads fail even when the model is told to.
- Pinned agent behaviour with an 18-case pytest evaluation suite covering cross-account isolation, source precedence, and a confirm-before-execute gate proving state-changing tools write nothing until confirmed.

Mistral-7B QLoRA Fine-Tuning

[GitHub](#) | [Model](#) | [Training logs](#)

Adapted Mistral-7B-Instruct-v0.3 on 500 domain instruction pairs, training 0.5% of parameters under NF4 4-bit quantization (r=64) and cutting perplexity 13% (cosine LR schedule, paged AdamW, gradient checkpointing).

Additional: [RAG system](#) (FastAPI, ChromaDB, LLaMA 3.1, cross-encoder re-ranking; live) | [CIFAR-10 benchmark](#) (92-97%)

EDUCATION

JECRC University – Master of Computer Applications (MCA), Computer Science, CGPA 8.57/10

Jul 2025 | Jaipur, India

MDS University – Bachelor of Computer Applications (BCA), Computer Science, CGPA 7.68/10

Jul 2023 | Ajmer, India

Certifications: [Claude 101](#) and [AI Fluency \(Anthropic\)](#) | [Responsible Generative AI \(Microsoft\)](#) | [Deep Learning \(KPMG\)](#)